

Supervised Learning – Conceptual Understanding

Part A: Theory | Brief Answers (2–4 lines each)

1. What are Supervised Learning Algorithms?

Supervised Learning algorithms learn from **labeled data**, where both input features and the correct target/output are provided. The model learns the relationship between them and uses it to predict the target for new, unseen data.

2. Difference between Regression Algorithms vs Classification Algorithms.

Regression predicts a continuous numerical value, such as house price, salary, or temperature.

Classification predicts a category or class, such as Yes/No, Spam/Not Spam, or Pass/Fail.

3. Explain Simple Linear Regression.

Simple Linear Regression predicts a continuous target using **one independent variable**. It fits a straight line to the data and estimates the target using the relationship between the input and output.

4. List and explain the Assumptions of Linear Regression.

Linearity: X and Y have a linear relationship. **Independence:** observations/errors are independent.

Homoscedasticity: error variance remains roughly constant. **Normality of residuals:** residuals are approximately normally distributed. **No strong multicollinearity:** independent variables should not be highly correlated with each other (mainly important in multiple regression).

5. What is Bias–Variance Trade-Off?

Bias is error caused by a model being too simple, while variance is error caused by a model being too sensitive to training data. The goal is to find a balance where the model is neither too simple nor unnecessarily complex, giving good performance on unseen data.

6. Explain Overfitting and Underfitting with examples.

Overfitting: the model learns the training data too closely, including noise, so training performance is high but test performance is poor. Example: a very complex polynomial model. **Underfitting:** the model is too simple to capture the pattern, so both training and test performance are poor. Example: using a straight line for a strongly curved relationship.

Quick Revision

Concept	Main Idea
Supervised Learning	Learn from labeled data
Regression	Predict a numerical value
Classification	Predict a category/class
Simple Linear Regression	One input → continuous output using a straight line
Bias–Variance	Balance simplicity and sensitivity
Overfitting	Good training, poor unseen-data performance

Underfitting	Poor training and poor unseen-data performance
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